



Assignment 01

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Section: B

Course Name: Cloud Computing

Topic: Git, Gitea, GitHub, Git LFS, and GitHub Pages

Task 1: Install Gitea and Push a Repository from the Ubuntu Server

Step 1.1: Install and Run Gitea on the Ubuntu Server

```
tayyabaziafat@ubuntu:~/Gitea$ docker compose ps
NAME                IMAGE              COMMAND                  SERVICE    CREATED        STATUS        PORTS
gitea               gitea/gitea:latest "/usr/bin/entrypoint..." gitea      16 minutes ago Up 16 minutes 22/tcp, 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
gitea_db            postgres:17-alpine "docker-entrypoint.s..." db         16 minutes ago Up 16 minutes 5432/tcp
tayyabaziafat@ubuntu:~/Gitea$ curl -sS -o /dev/null -w "Gitea HTTP status: %{http_code}\n" http://127.0.0.1:3000/
Gitea HTTP status: 200
tayyabaziafat@ubuntu:~/Gitea$ _
```

Step 1.2: Create and Push the Initial Repository

```
tayyabaziafat@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://tayyabaziafat@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 277 bytes | 69.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/tayyabaziafat/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
tayyabaziafat@ubuntu:~/Assignment01$ _
```

Step 1.3: Verify the Gitea Repository

The screenshot displays the Gitea web interface for a repository named 'Assignment01' owned by 'tayyabaziafat'. The interface is clean and modern, with a top navigation bar and a sidebar. The main content area shows the repository's commit history, which includes a single commit titled 'Add student information' made 20 minutes ago. Below the commit list, there is a file browser showing 'README.md'. The right sidebar provides additional information about the repository, including the number of stars (0), watchers (1), and forks (0). The footer of the page indicates that the interface is powered by Gitea version 28.0.0.

Task 2: Push the Same Repository to GitHub

Step 2.1: Create the GitHub Repository and Verify Both Remotes

```
tayyabaziafat@ubuntu:~/Assignment01$ git remote -v
gitea    http://tayyabaziafat@127.0.0.1:3000/tayyabaziafat/Assignment01.git (fetch)
gitea    http://tayyabaziafat@127.0.0.1:3000/tayyabaziafat/Assignment01.git (push)
github   https://github.com/tayyabaziafat/Assignment01.git (fetch)
github   https://github.com/tayyabaziafat/Assignment01.git (push)
tayyabaziafat@ubuntu:~/Assignment01$ _
```

Step 2.2: Push to GitHub

```
tayyabaziafat@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': tayyabaziafat
Password for 'https://tayyabaziafat@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 277 bytes | 55.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/tayyabaziafat/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
tayyabaziafat@ubuntu:~/Assignment01$ _
```

Step 2.3: Verify the GitHub Repository

The screenshot shows the GitHub web interface for a repository named 'Assignment01' owned by 'tayyabaziafat'. The repository is public and has 1 branch (main) and 0 tags. The commit history shows a single commit by 'tayyabaziafat' titled 'Add student information' from 11 hours ago. The README file is visible, containing the text 'Tayyaba Ziafat Registration Number:FA24B1-SE-018'. The right sidebar shows the repository's statistics: 0 stars, 0 watching, and 0 forks. The 'About' section indicates no description, website, or topics are provided. The 'Releases' and 'Packages' sections also show no published items.

Task 3: Track Three Large Files with Git LFS

Step 3.1: Configure Git LFS

```
tayyabaziafat@ubuntu:~/Assignment01$ cd ~/Assignment01
tayyabaziafat@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
tayyabaziafat@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
tayyabaziafat@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
tayyabaziafat@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
tayyabaziafat@ubuntu:~/Assignment01$
```

Step 3.2: Create and Verify Three LFS Files

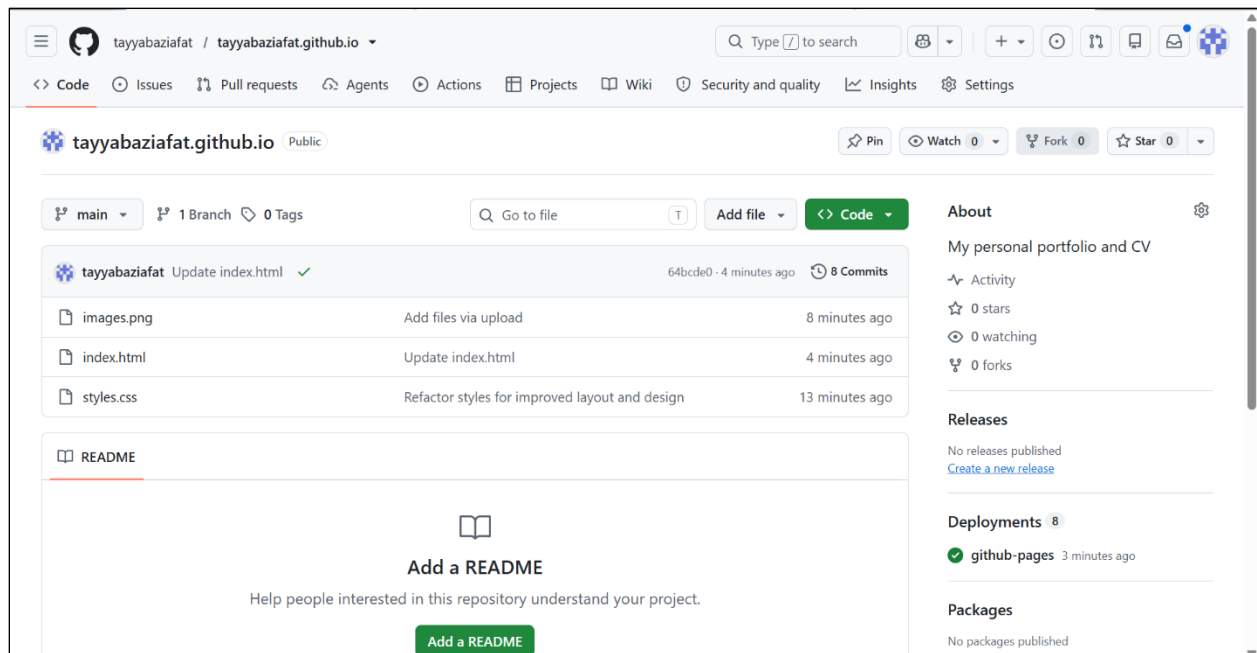
```
tayyabaziafat@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
tayyabaziafat@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
tayyabaziafat@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
tayyabaziafat@ubuntu:~/Assignment01$ git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
tayyabaziafat@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
tayyabaziafat@ubuntu:~/Assignment01$
```

Step 3.3: Commit and Push the LFS Files

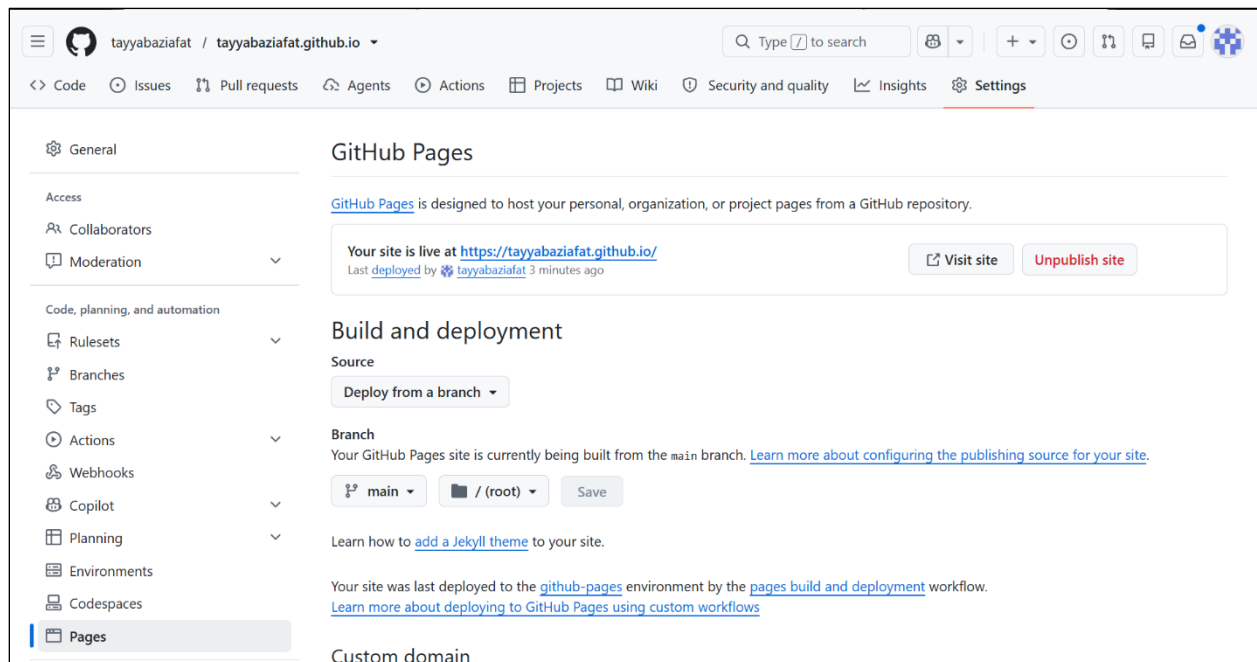
```
tayyabaziafat@ubuntu:~/Assignment01$ git push github main
Username for 'https://github.com': tayyabaziafat
Password for 'https://tayyabaziafat@github.com':
Username for 'https://github.com': tayyabaziafat
Password for 'https://tayyabaziafat@github.com':
Uploading LFS objects: 100% (3/3), 321 MB | 2.2 MB/s, done.
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 802 bytes | 267.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com:tayyabaziafat/Assignment01.git
 9b7d479..41c2841 main -> main
tayyabaziafat@ubuntu:~/Assignment01$ _
```

Task 4: Create a Portfolio or CV with GitHub Pages

Step 4.1: Create and Push the Portfolio Repository



Step 4.2: Publish with GitHub Pages



Step 4.3: Verify the Live Portfolio



